# Electromagnetic Induction

*Faraday's law, Lenz's law, inductance and the AC generator*

October 11, 2026

**Abstract**

A changing magnetic field can drive a current through a circuit with no battery at all. We follow that idea from Faraday's experiments to flux, Faraday's and Lenz's laws, inductance, RL circuits and the AC generator.

## 1 The experiments of Faraday and Henry

In 1831 Faraday and Henry found, independently, that a current appears in a closed coil whenever the magnetic field through it changes. A magnet, a coil and a galvanometer are enough to see it.

- A current flows only while the magnet and coil **move relative to each other**; at rest there is none, however strong the magnet.
- Faster motion gives a **larger** current, and reversing the motion **reverses** it.
- Switching the current in a *nearby* coil on or off has the same effect with no motion at all.

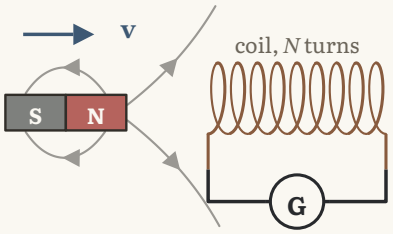



**Figure 1:** Moving the N pole in deflects G; moving it out reverses the deflection.

## 2 Magnetic flux

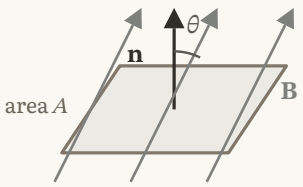



**Figure 2:** Only $B\cos\theta$ threads the area $A$.

Magnetic flux measures how much field threads a surface. It depends on the angle $\theta$ between **B** and the surface normal **n**: it is largest when the surface faces the field and zero when the field skims along it.

**Definition 1** (Magnetic flux).

Measured in webers, 1 Wb = 1 T m². For a curved surface, sum over every element $d\mathbf{A}$:

$$\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta, \qquad \Phi_B = \int \vec{B}\cdot d\vec{A} \tag{1}$$

## 3 Faraday's law of induction

**Faraday's law.**

The induced emf equals the negative rate of change of the flux linked with the circuit. Since $\Phi_B = BA\cos\theta$, it can come from changing $B$ (transformers), the area $A$ (sliding rods) or the angle $\theta$ (generators). For $N$ turns:

$$\boxed{\varepsilon = -N\,\frac{d\Phi_B}{dt}} \tag{2}$$

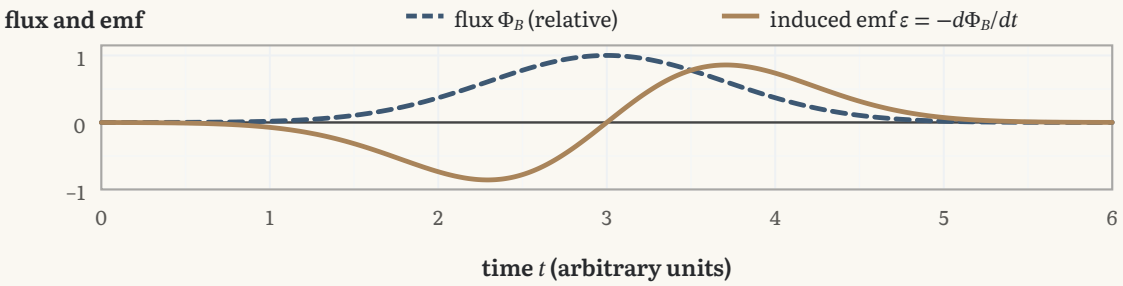



**Figure 3:** A magnet falling through a coil: the flux rises, peaks and falls, so the emf is a pulse that reverses where the flux peaks.

**Example 1** (A coil in a growing field).

A 200-turn coil of area 0.010 m² sees $B$ rise from 0.10 to 0.50 T in 0.20 s, so $|\varepsilon| = NA\,\Delta B/\Delta t = (200)(0.010)(0.40)/(0.20) = 4.0$ V. Through 10 Ω this drives $I = 0.40$ A.

## 4 Lenz's law: the direction of the current

**Lenz's law.**

The induced current opposes the change in flux that causes it. This is the minus sign in Faraday's law, and energy conservation in disguise.

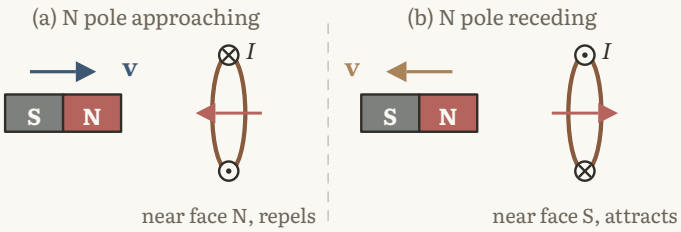



**Figure 4:** The induced field (red) opposes the magnet's while the flux grows (a) and aids it while the flux shrinks (b). Eddy currents in solid metal obey the same rule, which is how magnetic brakes work.

## 5 Motional emf

Flux also changes when the circuit changes shape. A rod of length $\ell$ slides at speed $v$ on rails joined by $R$; the enclosed area $\ell x$ gives $\Phi_B = B\ell x$.

$$\varepsilon = \frac{d\Phi_B}{dt} = B\ell\,\frac{dx}{dt} = B\ell v \tag{3}$$

The force $q\mathbf{v}\times\mathbf{B}$ drives charges up the rod. The current $I = B\ell v/R$ then feels a force $BI\ell$ opposing the motion, so the pushing power $Fv = \varepsilon^2/R$ is exactly the heat produced in $R$.

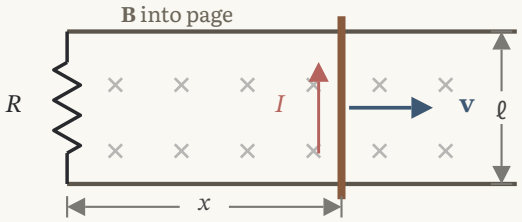



**Figure 5:** The sliding rod sweeps out area, so the flux grows at the rate $B\ell v$.

## 6 Self-inductance

A coil carrying current $I$ links its own flux $N\Phi_B = LI$, where $L$ is the **self-inductance**. The resulting *back emf* resists any change of current, giving the circuit electrical inertia:

$$\varepsilon = -L\,\frac{dI}{dt}, \qquad L_{\text{solenoid}} = \frac{\mu_0 N^2 A}{\ell} \quad \text{(henry, H)} \tag{4}$$

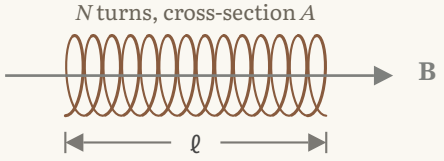



**Figure 6:** Inside a long solenoid, $B = \mu_0 n I$, $n = N/\ell$.

**Example 2** (Inductance of an air-cored coil).

With $N = 500$, $\ell = 25$ cm and $r = 1.0$ cm, $L \approx 0.39$ mH. An iron core multiplies this by $\mu_r$, often over a thousand.

## 7 Mutual inductance

A changing current in one coil (the primary) changes the flux through a neighbour (the secondary) and induces an emf there. This coupling is the principle of every transformer.

The secondary links flux $N_2\Phi_{21} = MI_1$, where $M$ is the **mutual inductance**. For coils on one long former, and in general,

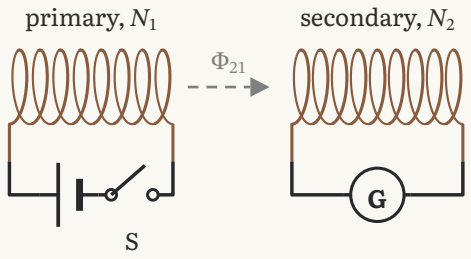



**Figure 7:** Switching $I_1$ on or off makes G kick in the separate secondary.

$$\varepsilon_2 = -M\,\frac{dI_1}{dt}, \qquad M = \frac{\mu_0 N_1 N_2 A}{\ell}, \qquad M = k\sqrt{L_1 L_2}\ \ (0 \le k \le 1) \tag{5}$$

## 8 Energy and the RL circuit

Building up current takes work against the back emf at the rate $LI\,dI/dt$. Integrating gives the stored energy, and its density for any magnetic field:

$$U = \int_0^I LI\,dI = \tfrac{1}{2}LI^2, \qquad u_B = \frac{B^2}{2\mu_0} \tag{6}$$

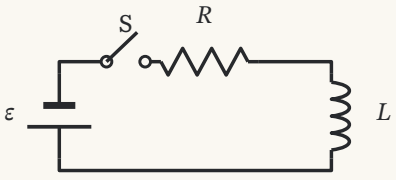



**Figure 8:** A series RL circuit.

The loop rule $\varepsilon = IR + L\,dI/dt$ gives an exponential rise; with the battery removed the current decays:

$$I_{\text{growth}} = \frac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right)$$
$$I_{\text{decay}} = I_0\,e^{-t/\tau}, \quad \tau = \frac{L}{R} \tag{7}$$

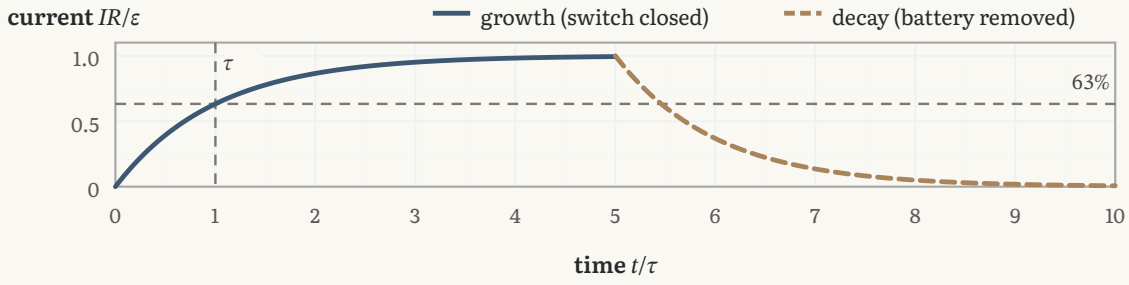



**Figure 9:** After one time constant the current reaches 63% of its final value; after $5\tau$, within 1%.

## 9 The AC generator

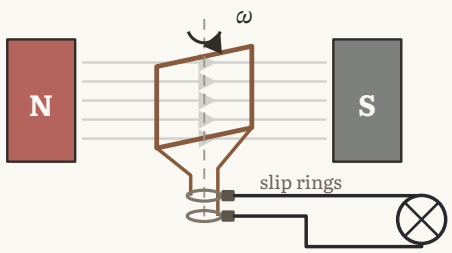



**Figure 10:** A simple AC generator.

A coil of $N$ turns and area $A$ rotating at angular speed $\omega$ in a uniform field has $\theta = \omega t$, so its flux varies as a cosine and the emf as a sine.

$$\Phi_B = BA\cos\omega t$$
$$\varepsilon = -N\frac{d\Phi_B}{dt} = NBA\omega\,\sin\omega t \tag{8}$$

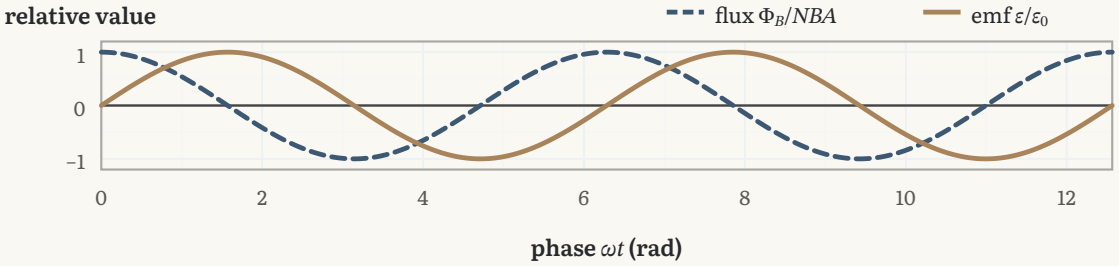



**Figure 11:** The emf peaks when the flux is zero, and vanishes when the flux peaks.

> **Example 3** (Mains frequency).
>
> 100 turns of area 0.050 m² turning at 50 Hz in 0.20 T: $\omega = 314$ rad s$^{-1}$, so $\varepsilon_0 = NBA\omega = 314$ V.

## 10 Summary

**Table 1:** The key results of the chapter.

| Quantity | Relation | SI unit |
|---|---|---|
| Magnetic flux | $\Phi_B = BA\cos\theta$ | weber (Wb) |
| Faraday's law | $\varepsilon = -N\,d\Phi_B/dt$ | volt (V) |
| Motional emf | $\varepsilon = B\ell v$ | volt (V) |
| Self-inductance | $\varepsilon = -L\,dI/dt,\ L = \mu_0 N^2 A/\ell$ | henry (H) |
| Mutual inductance | $\varepsilon_2 = -M\,dI_1/dt$ | henry (H) |
| Energy in an inductor | $U = LI^2/2$ | joule (J) |
| RL time constant | $\tau = L/R$ | second (s) |
| AC generator | $\varepsilon = NBA\omega\sin\omega t$ | volt (V) |